Exp 1 STUDY OF VARIOUS NETWORK COMMANDS USED IN LINUX & WINDOWS

1) arp a :- Show IP address of your computer along with address & MAC address of your router.

O/P

Interface 192.168.156.1 ... 0x7

Internet address	Physical address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-92	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-ff-ff-fa	static

2) Host Name :- Display name of your computer.

O/P

DESKTOP-MFFKGEN

3) Ipconfig/all : Display detailed configuration information about TCP/IP connection & Type of ethernet adapter in the system.

O/P :-

Windows IP Configuration

Host Name ... : DESKTOP-MFFKGEN

Primary DNS Suffix

Node type ... Mixed

IP routing enabled ... : NO

WINS proxy enabled ... : NO

Ethernet adapter Ethernet

Media state . . . : Media Disconnected

Connection specific DNS Suffix

Description . . . . : Realtek PCIe GbE Family Controller

Physical Address . . . : C4-D6-D3-A2-7Z-D3

DHCP enabled . . . : Yes

Autoconfiguration Enabled . . . . : Yes.

4) nbstat -a: Displays protocol statistics & current TCP/IP conditions using NBT

O/P

NBSTAT [[-a Remote Name] [-A IP address] [-c] [-n] [-N] [-R] [-RR] [-s] [Interval]]

-a (adapter status) -r (resolved) -RR (Relax/Refuse).

-A (Adapter status) -R (reload)

-c (cache) -s (sessions)

-n (names) -s (sessions)

5) netstat :- Display a variety of statics about a computer active TCP/IP connections.

O/P

Active Connections

Proto	Local address	Foreign address	state.
TCP	10.76.96.60.49175	10:76:96:163: domain	Time-wait
TCP	10.76.96.60:52239	142.250.182.3:http	ESTABLISHED
TCP	10:76.96.60.56959	10.76.96163:domain	Time wait

6) nslookup:- Display DNS Details such as IP address of the particular computer

O/P:-

server : Unknown

Address : 10.76.96.163

7) Pathping :- Traces the root of destination address then launches a 25 second test of each routed along the way

O/P:-

Pathping [-g host-list] [-h maximum-hops] [-i address]
[-n] [-p period] [-q num-queries] [-w timeout]
[-u] [-b] target_name.

LINUX NETWORKING COMMANDS.

1) ip:- Helps from setting up new system & assigning IPs to troubleshooting existing systems.

2) ipconfig :- The ipconfig command is a staple in many sys admin's tool kit for configuring & troubleshooting networks.

3) mtr:- MTR is a program with (L) that serves as a network diagnostic & troubleshooting tool.

4) tcpdump:- The tcpdump command is designed for capturing & displaying packets.

5) ping:- Ping is a tool that verifies IP level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP) Echo Request Message

1) ip → o/p

Usage: ip [OPTIONS] OBJECT {COMMAND / help}

ip [-force] -batch filename

where OBJECT = {link / address / addrlabel / route / rule / neigh / ntable / tunnel / tuntap / maddress / mroute / mud}

OPTIONS = {-V[ersion] [-s[tatistics] | -d[etails] | -r[esolve] -h[uman-readable] [-ies]}

2) ipconfig → o/p

lo: flags = 73<UP, LOOPBACK RUNNING> mtu 65536.

inet 127.0.0.1 netmask 255.0.0.0

inet 6::1 prefixlen 128 scopeid 0x10<host>

loop txqueuelen 1000 (Local Loopback).

RX errors 0 dropped 0 overruns 0 frame 0

TX packets 12329 bytes 871650 (792.16 KiB)

TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

3) mtr → o/p:-

My traceroute [v0.87]

localhost. localdomain (::1)

Keys: Help Display mode Restart Statistics Order of fields quit

Host	Packets			Pings			
	Loss%	Snt	Last	Avg	Best	Wrst	StDev
1.::1	0.0%	185	0.2	0.2	0.1	0.5	0.0

4) tcpdump o/p

tcpdump: enp3s0 You dont have permission to capture on device ....

Ping google.com

O/P

PING google.com (142.250.190.46) 56(84) bytes of data.

64 bytes from lga25s71-in-f14.1e100.net (142.250.190.46) icmp_seq=1 ttl=116

64 bytes from lga25s71-in-f14.1e100.net (142.250.190.46) icmp_seq=2 ttl=116.

^C

--- google.com ping statistics ---

2 packets transmitted, 2 received, 0% packet loss, time 300ms

RESULT:-

Thus the various command network used in linux & windows are executed successfully.

